

Nikita L. Polomoshnov

Faculty of Bioengineering and Bioinformatics, Lomonosov Moscow State University

V. N. Orekhovich Institute of Biomedical Chemistry, Moscow, Russia

nikitapol@fbb.msu.ru

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To the Editors

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Manuscript [ID], “Ground Truth That Does Not Ground: Substituting Reference Solvent σ -Profiles Degrades COSMO-SAC Solubility Prediction”, Polomoshnov & Rudik.

Dear Editors,

We write to report a correction to Section 3.3 of the above manuscript, which we found ourselves after submission. We would rather the referees read the section with this in hand than without it.

Section 3.3 reports the area the learned σ -profile places in the window COSMO-SAC’s hydrogen-bonding term reads, and concludes that the learned profile is chemically inadmissible there. The submitted text states that those areas were read off a retired COSMO-SAC head whose σ -supervision stream ran for zero steps, and it names the control it lacks: a supervised arm would separate an identifiability cause from an architectural one. We have since run that control, on the grounded arms that carry the substitution contrast itself (seeds 42 to 46). It moves the quantity by more than an order of magnitude. The median donor-window area, over the solvents whose reference tabulation is exactly zero, falls from 36.9 to 2.3 Å²; the median share of molecular surface falls from 31.0% to 1.5%; and the rank correlation between the learned and tabulated donor-window areas, across the thirty-one scored solvents, rises from -0.055 ($p = 0.77$) to $+0.65$ ($p = 8 \times 10^{-5}$).

The submitted conclusion therefore does not survive. Supervision recovers an ordering the unsupervised head did not have at all, and removes all but a twentieth of the mass. The window is not closed—the learned profile is occupied in every solvent whose reference is exactly zero—but “chemically inadmissible” overstates what is there. In the revision that paragraph and its figure are measured on the grounded arms, and the claim they carry is the one that survives: against the share of profile area the same window holds, the learned-minus-tabulated residual is over-represented there by a factor of 2.2 (IQR 1.6 to 3.0), exceeding one for 94% of molecules, quoted on the 48 whose reference tabulation carries donor area at all.

No other result is affected. The substitution contrast the paper reports, 1.93 to 2.34 in $\ln x_2$ MAE, is unchanged, as are the arm comparisons, the identifiability analysis and every number in the Supporting Information. The correction is contained in one paragraph, one figure and one caption.

We also flag an implication we do not yet resolve. Because the learned profile tracks the tabulated one more closely than the submitted Section 3.3 implies, the cost of substituting the tabulated profile is less readily attributed to a grossly inadmissible learned profile than that section suggests. We regard the mechanism as less settled than the submitted text reads, and the revision says so in those terms. The finding the paper reports—that substitution degrades prediction—rests on the measured contrast and is not affected.

The revised section, the regenerated figure and the analysis behind these numbers are prepared, and can be supplied now or at revision, whichever you prefer.

Yours sincerely,

Nikita L. Polomoshnov, on behalf of both authors